Real Applications Static Optimization

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These notes collect some *static* optimization problems that have appeared in various research papers. Students who take my Optimization class shall be able to solve and answer the questions listed in each model.

Your task:

1. Pick one model of your choice
2. Solve the model and derive the main results
3. Read the full paper or relevant excerpt or book chapter to understand the model and its application in the paper.
4. Prepare a short presentation slide, either in Vietnamese or English
   - The slides must be no longer than 10 pages
   - Font size must be 13pt or above
   - Each presentation is at most 15 minutes, with Q&A of 5 minutes.

1 Human capital and a poverty trap

Background. Parents invest in children's human capital under borrowing constraints or subsistence concerns; corner solutions at zero investment can generate poverty traps. Ceroni (2001) embeds this logic in a growth setting.[1]

Variables and parameters. c_t: parent's consumption; e_t: spending on the child's education (resource cost); h_t: child's human capital at the start of t (predetermined); h_{t+1}: human capital next period; A: productivity factor linking h_t to resources Ah_t; γ, v: scale and level in the human-capital technology; η: additive shifter in ln human capital; $\delta > 0$: weight on the child's future human capital in utility. All of $A, \gamma, v, \eta, \delta$ are parameters.

Objective.

$$\max_{c_t,\, e_t} \quad \ln c_t + \delta\, h_{t+1}.$$

Constraints.

- Flow resource constraint: $c_t = Ah_t - e_t$.
- Human capital technology: $h_{t+1} = \ln(\gamma e_t + v) + \eta$.
- Nonnegativity: $c_t \geq 0$, $e_t \geq 0$ (and implicitly $e_t \leq Ah_t$).

1. Show that substituting the constraints yields a problem in e_t only on $e_t \in [0, Ah_t]$, and that an *interior* optimum satisfies

$$\frac{1}{Ah_t - e_t} = \frac{\delta\gamma}{\gamma e_t + v}.$$

2. Show that there is a threshold $\underline{h} = v/(\delta\gamma A)$ such that $e_t^* = 0$ if $h_t \leq \underline{h}$ and, if $h_t > \underline{h}$,

$$e_t^* = \frac{\delta}{1+\delta} Ah_t - \frac{v}{\gamma(1+\delta)}.$$

Show that in the latter case $\dfrac{\partial e_t^*}{\partial h_t} > 0$.

[1] Carlotta Berti Ceroni (2001), "Poverty traps and human capital accumulation," *Economica* 68(270): 203–219.

2 Child quality and quantity

Background. Becker–Lewis quantity–quality tradeoffs explain why richer parents may have fewer children but invest more in each. De la Croix and Doepke (2003) link this mechanism to inequality and growth.[2]

Variables and parameters. c: parental consumption; $n > 0$: number of children (fertility); $e \geq 0$: education input per child; h: child "quality" (human capital); w: wage (market time endowment normalized); p: price of education; ϕ: time needed per child; $\theta \geq 0$: baseline human capital without parental spending; $\gamma \in (0,1)$: elasticity of quality in $e + \theta$; $\delta > 0$: taste for children. Parameters $w, p, \phi, \theta, \delta, \gamma$ as above.

Objective.

$$\max_{c,\,n,\,e} \quad \ln c + \delta \ln(nh), \qquad h = (e+\theta)^{\gamma}.$$

Constraints.

- Budget: $c + pen = (1 - \phi n)w$.
- Nonnegativity: $c > 0$, $n > 0$, $e \geq 0$.
- Feasibility of fertility: $\phi n < 1$ (mother has enough time).

1. Show that optimal consumption is always $c^* = w/(1+\delta)$. Show that if $w > \theta p/(\gamma\phi)$ then

$$e^* = \frac{\gamma\phi w/p - \theta}{1-\gamma}, \qquad n^* = \frac{\delta}{1+\delta}\,\frac{1-\gamma}{\phi w - \gamma p\theta},$$

whereas if $w \leq \theta p/(\gamma\phi)$ then $e^* = 0$ and $n^* = \dfrac{\delta}{1+\delta}\dfrac{1}{\phi}$.

2. Assume the interior regime $w > \theta p/(\gamma\phi)$. Show that

$$\frac{\partial n^*}{\partial w} < 0, \qquad \frac{\partial n^*}{\partial \phi} < 0.$$

Briefly interpret why a larger time cost ϕ lowers fertility.

[2] David de la Croix and Matthias Doepke (2003), "Inequality and growth: why differential fertility matters," *American Economic Review* 93(4): 1091–1113. See Gary S. Becker and Gregg H. Lewis (1973), "On the interaction between the quantity and quality of children," *Journal of Political Economy* 81(2, Part 2): S279–S288.

3 Gender wage gap and fertility

Background. When mothers supply childcare time, their wage enters the *marginal* cost of children. Galor and Weil (1996) use this to connect the gender gap to growth and fertility.[3]

Variables and parameters. c: couple's consumption; $n > 0$: fertility; w_m, w_f: husband's and wife's wages; ϕ: time per child borne by the wife; $\delta > 0$: taste for children. (The full model allows child quality; here $\gamma = 0$ in the quality term so only quantity enters utility.)

Objective.

$$\max_{c,\,n} \quad \ln c + \delta \ln n.$$

Constraints.

- Budget: $c + \phi w_f\, n = w_m + w_f$.
- Positivity: $c > 0$, $n > 0$.

1. Show that

$$c^* = \frac{w_m + w_f}{1+\delta}, \qquad n^* = \frac{\delta}{1+\delta}\,\frac{1}{\phi}\left(1 + \frac{w_m}{w_f}\right).$$

2. Show that

$$\frac{\partial n^*}{\partial w_f} < 0, \qquad \frac{\partial n^*}{\partial w_m} > 0.$$

If w_f rises toward w_m (narrowing the gender gap) holding w_m fixed, does fertility rise or fall?

[3] Oded Galor and David N. Weil (1996), "The gender gap, fertility, and growth," *American Economic Review* 86(3): 374–387. See Doepke et al. (2023, Sec. 2.4).

4 Marketization of childcare

Background. Outsourcing childcare turns part of the child-rearing cost from the mother's time into a money cost, flattening or reversing the fertility–female wage gradient (Ahn and Mira, 2002; Hazan and Zoabi, 2015).[4]

Variables and parameters. c, n: consumption and fertility; $s \in [0, \bar{s}]$: share of childcare bought on the market; p_s: monetary price per unit of outsourced childcare; ψ: goods cost per child; ϕ: total childcare time per child; w_m, w_f: wages; $\delta > 0$: taste for children.

Objective.

$$\max_{c,\,n} \quad \ln c + \delta \ln n.$$

Constraints. (for a fixed policy or equilibrium value of s)

- Budget:
$$c + \psi n + s p_s n \phi = w_m + w_f\big[1 - (1-s)n\phi\big].$$
- Positivity: $c > 0$, $n > 0$.

1. Show that

$$c^* = \frac{w_m + w_f}{1+\delta}, \qquad n^*(s) = \frac{\delta}{1+\delta}\,\frac{w_m + w_f}{\psi + \big[s p_s + (1-s) w_f\big]\phi}.$$

2. Show that

$$\frac{\partial n^*}{\partial w_f} = \frac{\delta}{1+\delta}\,\frac{\psi + \big(s p_s - (1-s) w_m\big)\phi}{\big(\psi + [s p_s + (1-s) w_f]\phi\big)^2}.$$

Assume $\psi < w_m \phi$. Show that $\partial n^*/\partial w_f < 0$ if and only if $s < \omega$, that $\partial n^*/\partial w_f = 0$ if $s = \omega$, and that $\partial n^*/\partial w_f > 0$ if $s > \omega$, where

$$\omega = \frac{w_m \phi - \psi}{(p_s + w_m)\phi}.$$

[4] Namkee Ahn and Pedro Mira (2002), "A note on the changing relationship between fertility and female employment rates in developed countries," *Journal of Population Economics* 15(4): 667–682. Moshe Hazan and Hosny Zoabi (2015), "Do highly educated women choose smaller families?" *Economic Journal* 125(587): 1191–1226. See Matthias Doepke *et al.* (2023, Sec. 4.2).

5 Careers and timing of fertility

Background. Delaying births can protect early-career investment but raises infertility risk. Doepke *et al.* (2023, Sec. 4.3) summarize this using a stylization related to Caucutt, Guner, and Knowles (2002).[5]

Variables and parameters. $e \in [0,1]$: time invested in career when young; $n_1, n_2 \in \{0,1\}$: child in period 1 or 2 (at most one child total); $c_1, c_{2,0}, c_{2,1}$: consumption when young, old without late child, old with late child; w_f: wage; κ, γ: scale and elasticity of second-period human capital $h(e) = \kappa e^\gamma$; ϕ: time cost of a child; $\pi \in (0,1)$: probability late birth succeeds; ν: utility from a child.

Objective. Given (n_1, n_2), maximize expected utility by choosing e:

$$\max_e \ \ln(c_1) + (1 - \pi n_2)\ln(c_{2,0}) + \pi n_2 \ln(c_{2,1}) + \nu n_1 + \pi \nu n_2.$$

Constraints.

- $c_1 = w_f(1 - n_1\phi - e)$.
- If $n_2 = 0$: second-period consumption is $c_{2,0} = w_f \kappa e^\gamma$.
- If $n_2 = 1$: $c_{2,1} = w_f \kappa e^\gamma (1-\phi)$ when birth succeeds (and $c_{2,0}$ applies with probability $1-\pi$ as in the objective).

1. Show that, given n_1, n_2, the optimal career choice is

$$e = \frac{\gamma}{1+\gamma}(1 - n_1\phi).$$

2. Take $n_1 = n_2 = 0$ (no birth at all), show that the optimal utility is

$$u_{0,0} = \ln\left(\frac{w_f}{1+\gamma}\right) + \ln\left(\kappa w_f \left(\frac{\gamma}{1+\gamma}\right)^\gamma\right)$$

3. Take $n_1 = 1, n_2 = 0$ (no late birth), so second-period utility is $\ln(c_{2,0})$ with $c_{2,0} = w_f\kappa e^\gamma$. Show that the optimal utility is

$$u_{1,0} = u_{0,0} + (1+\gamma)\ln(1-\phi) + \nu$$

4. Take $n_1 = 0, n_2 = 1$, show that the optimal utility is

$$u_{0,1} = u_{0,0} + \pi[\ln(1-\phi) + \nu].$$

5. Under what condition of ν does having children early rather than late yield higher utility to the women?

[5] Elizabeth M. Caucutt, Nezih Guner, and John Knowles (2002), "Why do women wait? Matching, wage inequality, and the incentives for fertility delay," *Review of Economic Dynamics* 5(4): 815–855. Claudia Goldin (2014), "A grand gender convergence: its last chapter," *American Economic Review* 104(4): 1091–1119.

6 Nash bargaining on a surplus

Background. Cooperative Nash bargaining delivers a Pareto efficient split of surplus given a threat point; it underpins wage and household models (Nash, 1950; Manser and Brown, 1980; McElroy and Horney, 1981).[6]

Variables and parameters. π_1, π_2: bargained payoffs on the Pareto frontier; d_1, d_2: disagreement (threat-point) utilities; total feasible utility normalized to $\pi_1 + \pi_2 = 1$.

Objective.

$$\max_{\pi_1} \quad (\pi_1 - d_1)(1 - \pi_1 - d_2).$$

Constraints.

- Feasible frontier: $\pi_2 = 1 - \pi_1$.
- Individual rationality (for now assume interior): $\pi_1 \geq d_1$, $\pi_2 \geq d_2$.
- Feasible surplus: $d_1 + d_2 < 1$.

1. Show that the unconstrained maximizer is

$$\pi_1^* = \tfrac{1}{2}(1 + d_1 - d_2).$$

2. Show that $\partial \pi_1^* / \partial d_1 = \frac{1}{2}$ and $\partial \pi_1^* / \partial d_2 = -\frac{1}{2}$.

Extension. We now extend the previous model to account for asymmetric bargaining power. Bargaining power weights enter macro and family applications (e.g. Chiappori, 1992; Muthoo, 1999).[7]

Variables and parameters. $\alpha \in (0, 1)$: weight on agent 1's surplus in the generalized Nash product; d_1, d_2, π_1, π_2 as above.

Objective.

$$\max_{\pi_1} \quad (\pi_1 - d_1)^{\alpha}(1 - \pi_1 - d_2)^{1-\alpha}.$$

Constraints. Same as the previous section (interior participation, $d_1 + d_2 < 1$).

1. Show that

$$\pi_1^* = d_1 + \alpha(1 - d_1 - d_2), \qquad \pi_2^* = d_2 + (1 - \alpha)(1 - d_1 - d_2).$$

2. Show that $\partial \pi_1^* / \partial \alpha = 1 - d_1 - d_2 > 0$ when surplus is positive. Interpret α as the bargaining power of agent 1.

[6] John F. Nash, Jr. (1950), "The bargaining problem," *Econometrica* 18(2): 155–162. Marilyn Manser and Murray Brown (1980), "Marriage and household decision-making: a bargaining analysis," *International Economic Review* 21(1): 31–44. Marjorie B. McElroy and Mary Jean Horney (1981), "Nash-bargained household decisions: toward a generalization of the theory of demand," *International Economic Review* 22(2): 333–349.

[7] Pierre-André Chiappori (1992), "Collective labor supply and welfare," *Journal of Political Economy* 100(3): 437–467. Abhinay Muthoo (1999), *Bargaining Theory with Applications*, Cambridge University Press.

7 Voluntary childlessness

Background. Childlessness can reflect optimal corners when fixed costs of children (money or utility) are large relative to taste for children; see Baudin *et al.* (2019) and Doepke *et al.* (2023, Sec. 6.1).[8]

Variables and parameters. c: consumption; $n \in \{0, 1\}$: no child / one child; w: full income; $k \geq 0$: money cost of a child; $\delta > 0$: utility from a child; $F > 0$: additive utility (psychic) cost of a child.

Objective. Compare utility at $n = 0$ versus $n = 1$:

$$U = \ln(c) + \delta n - Fn.$$

Constraints.

- Budget: $c + kn = w$.
- Integer fertility: $n \in \{0, 1\}$.
- Feasibility: $w > k$ if $n = 1$ is to be affordable.

1. Show that $n = 0$ is optimal if and only if

$$F \geq \delta + \ln\left(\frac{w - k}{w}\right) \qquad \text{(for } w > k\text{)}.$$

2. Define $\Delta(w) \equiv U(n = 1) - U(n = 0) = \ln(w - k) + \delta - F - \ln w$. Show that for $w > k$,

$$\Delta'(w) = \frac{1}{w - k} - \frac{1}{w} = \frac{k}{w(w - k)} > 0.$$

Interpret: richer households are more likely to afford/tolerate a child when $k > 0$.

[8] Thomas Baudin, David de la Croix, and Paula E. Gobbi (2019), "Childlessness and economic development," in *Human Capital and Economic Growth*, Springer, 55–94.

8 Social norms and outsourced childcare

Background. Deviating from a social norm on market childcare is psychologically costly; this can interact with wages (Doepke *et al.*, 2023, Sec. 5.3; Hazan *et al.*, 2021).[9]

Variables and parameters. $s \in [0,1]$: share of childcare purchased; p_s: price of outsourced childcare; ϕ: total childcare time per child; w_m, w_f: wages; $\tau > 0$: strength of the norm; $s^* \in [0,1]$: reference (societal) level of s. (Fix $n = 1$ child.)

Objective. With one child, indirect utility as a function of s is

$$\max_{s\in[0,1]} \quad \underbrace{w_m + w_f\big(1-(1-s)\phi\big) - sp_s\phi}_{\text{full income minus childcare spending}} - \frac{\tau}{2}(s-s^*)^2.$$

Constraints.

- Choice set: $s \in [0,1]$.
- Parameters: $\tau > 0$, $p_s, \phi, w_m, w_f > 0$, $s^* \in [0,1]$.

1. Show that an unconstrained interior critical point satisfies

$$\hat{s} = s^* + \frac{\phi\,(w_f - p_s)}{\tau}.$$

State when the optimum is $s = 0$ or $s = 1$ (corners) instead of $\hat{s}$.

2. For an interior solution $\hat{s} \in (0,1)$, show that

$$\frac{\partial \hat{s}}{\partial w_f} = \frac{\phi}{\tau} > 0, \qquad \frac{\partial \hat{s}}{\partial \tau} = -\frac{\phi\,(w_f - p_s)}{\tau^2}.$$

Explain how larger τ pulls chosen s toward s^*.

[9] Matthias Doepke *et al.* (2023, Sec. 5.3). Moshe Hazan, David Weiss, and Hosny Zoabi (2021), "An aggregate model of changing sectoral and occupational female employment in the US: 1960–2010," CESifo Working Paper No. 8803; the fertility handbook chapter also cites Hazan *et al.* (2021a) on marketization and childlessness among highly educated women.

9 Samuelson: two-period consumption loan

Background. Samuelson (1958) launched the OLG toolkit used in public finance and macro.[10]

Variables and parameters. c_1, c_2: consumption when young and old; ω_1, ω_2: endowments; $R > 1$: gross interest factor (price of future consumption in present value); $\beta > 0$: discount factor on old-age utility.

Objective.

$$\max_{c_1,\, c_2} \quad \ln c_1 + \beta \ln c_2.$$

Constraints.

- Lifetime budget: $c_1 + \dfrac{c_2}{R} = \omega_1 + \dfrac{\omega_2}{R} \equiv W$.
- Positivity: $c_1, c_2 > 0$.

1. Show that, with $W = \omega_1 + \omega_2/R$,

$$c_1^* = \frac{W}{1+\beta}, \qquad c_2^* = \frac{\beta R W}{1+\beta}.$$

2. Show that

$$\frac{\partial c_1^*}{\partial R} = \frac{1}{1+\beta}\left(-\frac{\omega_2}{R^2}\right), \qquad \frac{\partial c_2^*}{\partial R} = \frac{\beta W}{1+\beta} + \frac{\beta R}{1+\beta}\left(-\frac{\omega_2}{R^2}\right).$$

If $\omega_2 > 0$, does c_1^* rise or fall when R increases? (Income vs. substitution: c_2^* has two terms.)

3. Now, the government imposes an income tax τ on citizens so that the new budget constraint becomes

$$c_1 + \frac{c_2}{R} = (1-\tau)W$$

and the government reimburses the collected tax on improving citizen welfare by providing public goods that are free to obtain, so that the citizens' new welfare becomes

$$\ln c_1 + \beta \ln c_2 + \theta G$$

Citizens take τ as given and maximize utility. Under what condition is the tax τ desirable?

[10] Paul A. Samuelson (1958), "An exact consumption-loan model of interest with or without the social contrivance of money," *Journal of Political Economy* 66(6): 467–482.

10 Fehr–Ujhelyiova: static model of fertility choice

Background. Before their overlapping-generations simulation, Fehr and Ujhelyiova (2013, Sec. 2) motivate the quantitative analysis with a stripped-down static problem: a household maximizes separable CES utility over consumption c and fertility n, subject to a budget in which children entail a money cost b_0 per child and a time cost b_1 per child while the woman's time endowment is normalized to one and the wage is w.[11]

Variables and parameters. $c > 0$: consumption; $n > 0$: number of children; $a_c, a_n > 0$: taste weights on c and n; $\rho > 0$, $\rho \neq 1$: elasticity of substitution in the CES subutilities; $w > 0$: wage (full income when all time is worked); $b_0 \geq 0$: monetary cost of a child; $b_1 \geq 0$: time cost of a child (fraction of the unit time endowment).

Objective. (Fehr and Ujhelyiova 2013, p. 4)

$$\max_{c,\,n} \quad U(c,n) = a_c\,\frac{c^{1-1/\rho}}{1-1/\rho} \;+\; a_n\,\frac{n^{1-1/\rho}}{1-1/\rho}.$$

Constraints.

- Budget (time for work is $1 - b_1 n$, so labor income is $w(1 - b_1 n)$; children cost $b_0 n$ in goods):
$$c + b_0 n = w(1 - b_1 n).$$
- Equivalently, full income spent on consumption at price one and on children at shadow price $b_0 + wb_1$:
$$c + (b_0 + wb_1)\,n = w.$$
- Interiority: $c > 0$, $n > 0$, and $1 - b_1 n > 0$.

1. Define the Lagrangian with multiplier λ on $w - c - (b_0 + wb_1)n$. Show that interior first-order conditions imply

$$a_c\,c^{-1/\rho} = \lambda, \qquad a_n\,n^{-1/\rho} = \lambda\,(b_0 + wb_1).$$

2. Combine the two FOCs to obtain $(a_n/a_c)(c/n)^{1/\rho} = b_0 + wb_1$, hence $c = n\,\big[(a_c/a_n)(b_0 + wb_1)\big]^{\rho}$. Substitute into $c + (b_0 + wb_1)n = w$ and show that

$$n^* \;=\; \frac{w}{\left(\dfrac{a_c}{a_n}(b_0 + wb_1)\right)^{\rho} \;+\; b_0 + wb_1}\,.$$

(Fehr and Ujhelyiova print the same denominator with the first term written as $\frac{a_c}{a_n}(b_0 + wb_1)^{\rho}$, i.e. with the power on the price term only; the expression above is the one implied by their two FOCs.)

3. Let $p \equiv b_0 + wb_1$. Treating w as fixed, show that $\partial n^*/\partial b_0 < 0$ and $\partial n^*/\partial b_1 < 0$ (higher money or time cost of children lowers fertility). Relate sign shifts to policies that reduce b_0 or b_1.

[11] Hans Fehr and Daniela Ujhelyiova (2013), "Fertility, female labor supply, and family policy," *German Economic Review* 14(2): 138–165, `doi:10.1111/j.1468-0475.2012.00568.x`. CESifo Working Paper No. 3455 (2011). They interpret policy as lowering b_0 (e.g. transfers) or b_1 (e.g. subsidized child care).